Keystone

Deterministic Replay-Verifiable Trust Pipeline

Keystone is a deterministic replay-verifiable trust pipeline for proving software execution integrity across multi-stage repository workflows. It is not positioned as an AI startup pitch, productivity tool, or CI wrapper. It is proof infrastructure for deterministic execution, canonical artifact generation, immutable stage lineage, and replay-safe verification.

Validated Proof Status

Target	Evidence Produced
Keystone self-verification	Two SAFE scans, identical proof hash, byte-identical proof folder comparison, replay SAFE.
Clean external repository	Repeated proof execution with identical final hash and replay SAFE verification.
Messy real-world repository	Actively modified repository validated with identical proof hash and replay SAFE verification.

Most Important Proof Claims

- Same repository input produces the same final proof hash across repeated executions.
- Replay verifies the trusted run and confirms the proof chain has not silently changed.
- Keystone successfully verifies Keystone itself.
- Independent repository validation is included alongside Keystone self-verification.
- Canonicalized artifact integrity is represented through manifests, hashes, and replayable command evidence.

Positioning

Keystone is positioned as deterministic orchestration, replay-verifiable execution, canonical artifact proof, fail-closed verification, and AI-generated code trust infrastructure.

Core Capabilities

- Deterministic multi-stage execution
- Canonical artifact generation
- Cryptographic stage manifests
- Immutable proof lineage
- Replay verification across all stages
- Fail-closed trust enforcement
- Byte-identical repeated-run proof
- External and messy repository validation

Enterprise Relevance

- Software supply-chain integrity
- Reproducible infrastructure verification
- Deterministic AI-generated code governance
- Compliance-grade execution evidence
- Tamper-evident pipeline history

Proof Package Contents

Demo video: 03_Demo/keystone_deterministic_demo.mp4

Architecture diagram: 02_Architecture/keystone_deterministic_trust_pipeline.png

Proof artifacts: 04_Proof_Artifacts/ full-resolution screenshots

Replay commands: 06_Replay_Commands/deterministic_replay_commands.txt

Package integrity: MANIFEST_SHA256.json

Architecture